

HASSAN JAMAL

hjamal.bs24seecs@seecs.edu.pk — 0328-9082754 — LinkedIn — Portfolio — GitHub

Education

National University of Sciences and Technology (NUST)
BS Computer Science — 4th Semester

2024–Present

Skills

Languages: Python, C++, Java, JavaScript, PL/SQL	AI Tools: Antigravity, Gen-AI / LLM (Groq, Gemini),
Backend: Node.js, Express.js, MongoDB, REST APIs,	Cursor, GitHub Copilot, Kiro, Lovable
JWT Auth, CRUD, MVC, WebContainer API, WebSockets	Tools: Git, GitHub, VS Code, PyCharm, Postman, Figma
(Socket.IO)	Core: DSA, OOP, DB Systems, Networking, Real-Time
Frontend: React.js, HTML, CSS, Tailwind CSS, Bootstrap	Systems, CRDTs, REST APIs
Cloud & DevOps: Docker, AWS (ECR, ECS, ALB), Nginx	

Projects

FREYA : AI-Powered Collaborative Code Editor (Gen-AI + MERN + WebContainers)

Live Demo GitHub

- Built a full-stack Gen-AI collaborative development environment enabling real-time multi-user coding, editing, and execution directly in the browser with live synchronization and project management.
- Integrated **Groq LLM** for AI-powered code generation and modification, implemented Socket.io-based real-time collaboration with automatic code-to-file conversion, and used **WebContainer API** to run Node.js/React apps without local setup.

UrbanPulse: GeoSpatial Urban Intelligence Platform (MERN Stack)

GitHub

- Built a full-stack MERN-based geospatial analysis system using OpenStreetMap and Overpass APIs.
- Developed backend services for spatial data processing along with infrastructure scoring and route analysis modules.

Plantora - Full-Stack Plant E-Commerce Platform (MERN Stack)

Live Demo GitHub

- Built a production-ready full-stack plant e-commerce platform using the **MERN stack** with secure JWT authentication, role-based admin access, persistent cart synchronization, and real-time order management.
- Developed advanced e-commerce workflows including dynamic plant catalog filtering, animated 3D featured plant carousel, Cloudinary-based media management, receipt upload verification, and hybrid cart persistence using MongoDB for authenticated users and localStorage for guests.

Gapify AI - Resume & Interview Analyzer (Gen-AI + MERN Stack)

Live Demo GitHub

- Built a full-stack Gen-AI powered MERN application that analyzes resumes and job descriptions to generate match scores, skill gaps, interview questions, and preparation plans.
- Integrated **Groq LLM API** for intelligent analysis and used **MongoDB** for authentication, user data storage, and report management, deployed on Vercel (frontend) and Render (backend).

Encrypted VPN with Login System

GitHub

- Developed a secure VPN system with encrypted client-server communication and authentication mechanisms.
- Implemented session management and low-level packet handling using a custom network architecture.

Dimensional Aspect-Based Sentiment Analysis (SemEval-2026)

GitHub

- Developed an NLP-based sentiment analysis system using deep learning for aspect-level classification.
- Performed data preprocessing, feature engineering, and model optimization to improve accuracy.

Recommendation System with Graph Analysis

GitHub

- Built a graph-based recommendation engine using user-item interaction networks.
- Applied network analysis techniques to extract patterns and improve recommendation quality.

Chess Arena - Real-Time Multiplayer Game (Docker + Render Deployment)

GitHub Live Demo

- Built a real-time multiplayer chess game with live chat using **WebSockets** for instant synchronization across devices.
- Implemented player roles and game logic using **Socket.IO**.
- Containerized with **Docker** and deployed on Render, debugging production WebSocket issues.